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3.6.2 Types of Stem

Majority of angiosperm possess upright, vertically growing erect stem. They are (i) Excurrent, (ii) Decurrent, (iii) Caudex, (iv) Culm.

i. Excurrent

The main axis shows continuous growth and the lateral branches gradually becoming shorter towards the apex which gives a conical appearance to the trees. Example: *Polyalthia longifolia, Casuarina.*

ii. Decurrent

The growth of lateral branch is more vigorous than that of main axis. The tree has a rounded or spreading appearance. Example: *Mangifera indica, Azadirachta indica, Tamarindus indicus, Aegle marmelos*

iii. Caudex

It's an unbranched, stout, cylindrical stem, marked with scars of fallen leaves. Example: *Cocus nucifera, Borassus flabelliformis, Areca catechu*

iv. Culm

Erect stems with distinct nodes and usually hollow internodes clasped by leaf sheaths. Example: Majority of grasses including Bamboo.

3.6.3 Modification of Stem

I. Aerial modification of stem

1. Creepers

These are plants growing closer (horizontally) to the ground and produces roots at each node. Example: *Cynodon dactylon, Oxalis, Centella*

2. Trailers (Stragglers)

It is a weak stem that spreads over the surface of the ground without rooting at nodes. They are divided into 3 types,

i. **Prostrate (Procumbent):** A stem that grows flat on the ground. Example: *Evolvulus alsinoides, Indigofera prostrata.*

ii. **Decumbent:** A stem that grows flat but becomes erect during reproductive stage. Example: *Portulaca, Tridax, Lindenbergia*

iii. **Diffuse:** A trailing stem with spreading branches. Example: *Boerhaavia diffusa, Merremia tridentata*

3. Climbers

These plants have long weak stem and produce special organs for attachment for climbing over a support. Climbing helps to display the leaves towards sunlight and to position the flower for effective pollination.

i. Root climbers

Plants climbing with the help of adventitious roots (arise from nodes) as in species of *Piper betel, Piper nigrum, Hedera helix, Pothos, Hoya.*

ii. Stem climbers (twiners)

These climbers lack specialised structure for climbing and the stem itself coils around the support. Example: *Ipomoea, Convolvulus, Dolichos, Clitoria, Quisqualis.*

Stem climbers may coil around the support clockwise or anti-clockwise. Clockwise coiling climbers are called **dextrose.** Example: *Dioscorea alata*. Anti-clockwise coiling climbers are called **sinistrose.** Example: *Dioscorea bulbifera*.